THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Sound cannot travel in a vacuum. Reason (R): Sound waves need a material medium for propagation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A sound wave travels from air (340 m/s) into water (1500 m/s). A 500 Hz wave has wavelengths in air and water of: 1

(a) 3 m and 0.68 m **(b)** 0.68 m and 0.68 m
(c) 3 m and 3 m **(d)** 0.68 m and 3 m

3. If the temperature rises from 0 degrees Celsius to 40 degrees Celsius, the speed of sound in air increases from 331 m/s by about (0.6 m/s per degree Celsius): 1

(a) 40 m/s **(b)** 240 m/s **(c)** 24 m/s **(d)** 6 m/s

4. A sonar signal gives echoes from two surfaces at 0.8 s and 1.0 s. The separation between the surfaces (v = 1500 m/s) is: 1

(a) 600 m **(b)** 75 m **(c)** 300 m **(d)** 150 m

5. Assertion (A): The speed of sound is greater in solids than in gases. Reason (R): The particles in solids are closely packed and strongly bound. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A person standing at 340 m from a cliff fires a gun. The echo is heard after: 1

(a) 1 s **(b)** 0.5 s **(c)** 4 s **(d)** 2 s

7. A sound of 20 degrees Celsius air moves at 344 m/s. The distance travelled in 0.25 s is: 1

(a) 86 m **(b)** 43 m **(c)** 1376 m **(d)** 172 m

8. A pulse takes 0.1 s to travel along a 34 m long steel pipe. The speed in steel is: 1

(a) 340 m/s **(b)** 34 m/s **(c)** 0.0029 m/s **(d)** 3400 m/s

9. Assertion (A): A louder sound has a greater frequency. Reason (R): Loudness depends on the amplitude only. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. A person hears the echo of a gun shot 1 s after the shot and a second echo 3 s after. The ratio of the distances of the two reflectors is: 1

(a) 3 : 1 **(b)** 1 : 2 **(c)** 1 : 3 **(d)** 1 : 1

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A boy standing 85 m in front of a hill claps and hears the echo after 0.5 s. Find the speed of sound. 2

12. Why are two sounds of the same loudness and pitch from a violin and a sitar distinguishable? 2

13. State why the sound of an approaching train is first heard through the rails. 2

14. The speed of sound in steel is 5000 m/s. A worker strikes one end of a 1 km steel rail. Find the time interval between the sounds heard through steel and air (v = 340 m/s) at the other end. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two sound waves have frequencies 100 Hz and 400 Hz with equal amplitude. Compare their wavelengths, speeds, loudness and pitch in the same medium. 3

16. A siren emits a sound of frequency 700 Hz. Find the time period, the wavelength in air (340 m/s), and the number of waves reaching a man in 5 s. 3

17. Explain why the sound of a bell is heard for a long time in a stone temple but only for a short time in an open field. 3

OR

A man fires a gun in front of a cliff and hears an echo after 2 s and a second after 3 s from another cliff. Find the distance between the cliffs if he stands between them (v = 340 m/s). 3

18. A man claps near a big cliff and hears the echo after 3 s. He then walks 85 m toward the cliff and claps again. Find the new echo time (v = 340 m/s). 3

OR

Describe the working of an electric bell in a vacuum jar and a microphone as a transducer. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A source of frequency 440 Hz sends a sound wave in air (344 m/s) towards a wall 172 m away. Find the wavelength, the time for the echo, the number of waves between the source and the wall, and the number of complete waves in the echo path. 5

OR

Explain why echoes cannot be heard in a small room, but multiple echoes are heard in a big hall, and design the dimensions needed for a hall so that reverberation time is reduced. 5

20. Describe an experiment to measure the speed of sound in air using two observers and a gun or a clapper at a large distance. State the observations, the calculation for 1020 m with a time of 3 s, and the precautions. 5

OR

Describe how ultrasound is used in the SONAR, in medical sonography and in non-destructive testing, including any limitations. 5

END OF QUESTION PAPER — Total: 40 Marks